

ZeroDrift Mk3 — what to buy, and from where

Total ₹6,345 · ✓ = price verified against a live listing, ~ = estimate · generated from [docs/data/bom_tier_a.json](#)

ORDER THE UNDERLINED ITEMS TONIGHT. I could not see Amar Robotics' actual stock — their Instagram and IndiaMart pages are blocked from where I work — so this is routed by component *type*. The underlined five are specialist parts that a general electronics shop and a 10-minute grocery app both almost never carry, and **three of them stop the build dead if they are missing**. They are also the only items with real delivery lead time. The shop trip can happen any day; the order cannot be compressed.

1 · ORDER ONLINE NOW — ₹2,377

Amazon.in / Robu.in / ThinkRobotics. Not a local-shop item. If the shop *does* have the AS5600 or TCA9548A, buy them there and cancel that line — but ask whether the AS5600 includes a magnet, and whether it is **diametrically** magnetised. Most ship bare or with the wrong type, and an axial magnet looks identical and does not work.

2	<u>AS5600 magnetic encoder module</u>	✓	₹498
2	<u>Diametric magnet, 6 × 2.5 mm</u>	~	₹180
1	<u>TCA9548A I²C multiplexer</u>	✓	₹199
1	<u>USB camera module, M12 mount, 12 mm lens</u>	~	₹1,400
1	<u>Red filter — gel sheet or red acrylic offcut</u>	~	₹100
subtotal			₹2,377

2 · AMAR ROBOTICS / LOCAL SHOP — ₹3,839

Phone first for the four motion parts at the top — stocked by some shops, not all. Everything below them is standard hobby stock. The 3×AA holder is the decoy's supply — see the power note.

2	NEMA17 stepper motor, 1.8°/step, 4.2 kg-cm	✓	₹1,498
2	A4988 stepper driver module	✓	₹340
2	NEMA17 L-bracket (pan + tilt)	✓	₹254
2	100 µF 25 V electrolytic capacitor	~	₹20
1	ABS project box, head enclosure	~	₹115
1	Arduino Nano — beacon	~	₹225
1	12 V 2 A DC power adapter	✓	₹279
1	Acrylic sheet 3 mm, 6"×6" (pack of 2)	✓	₹199
1	Vibration motor, 3 V DC	✓	₹20
1	Transistor + diode + resistor combo kit	✓	₹105
1	High-brightness RED LED, 10 mm, 650 nm	~	₹22
1	2N2222 transistor + 220 Ω + 1 kΩ	~	₹20
1	SPST toggle switch	~	₹25
1	ABS project box	~	₹115
1	High-brightness WHITE LED (brighter than beacon)	~	₹22
1	68 Ω resistor	~	₹5
1	3×AA battery holder + 3 AA cells — decoy supply	~	₹90
1	Switch + scrap housing	~	₹35
1	Dupont jumper wires — M-M, M-F, F-F	~	₹150
1	M3 screws, nuts, standoffs assortment	~	₹150
1	Heat-shrink, solder, hot-glue sticks	~	₹150
subtotal			₹3,839

3 · BLINKIT / ANY KIRANA — ₹129

Not electronics. A sports shop sells these for about a quarter of the quick-commerce price — only use Blinkit if the ten minutes matter more than ₹100.

1	Ping-pong ball (diffuser)	✓	₹129
subtotal			₹129

4 · ALREADY OWNED — ₹0

No 9 V battery and no coin cell — see the power note.

—	KY-008 laser module	—	₹0
—	Arduino Nano — tracker	—	₹0
—	USB power bank (any, 5V) — beacon supply	—	₹0

POWER: WHY NOT 9V, WHY NOT A COIN CELL, AND WHY 3×AA FOR THE DECOY

The tracker Nano runs off the laptop's USB and the motors off the 12 V adapter — neither needs a battery. The beacon runs off a **USB power bank** you already own: a 9 V alkaline holds ~500 mAh and the Nano burns the 9→5 V drop as heat, so the bank is both cheaper and better. The decoy is the one that needed real arithmetic, not a guess: a white LED's forward voltage (~3.1 V) is at or above what a CR2032 *or* 2×AA can supply, so at either one the LED dims to nothing within hours as the battery sags — the opposite of "brighter than the beacon," which is the whole point of the decoy. **3×AA (4.5 V) with a 68 Ω resistor** (not 150 Ω) draws 16–25 mA across the pack's full working life — genuinely bright throughout, not just on day one.

CHECK AT THE COUNTER

- Magnet must be DIAMETRIC**, not axial. Bring a second small magnet: a diametric one attracts on its curved *side* edge, an axial one on its flat faces.
- Look at the motor's rear**. A short shaft out the back = dual-shaft, encoder mounts cleanly. Flat back is fine, but the sensor then needs a short, stiff overhanging arm.
- Capacitors 25 V or higher**. 16 V on a 12 V rail has no margin against the inductive kick they exist to absorb — and skipping them kills both drivers on first power-up.
- Camera must allow manual exposure (UVC)**. Auto-exposure hunts on every blink, and the blink is the signal. Ask for 12 mm; 8 mm acceptable, 6 mm at a push.